

# Real-Time Human–Computer Interaction Using CNN-Based Gaze Tracking for Hands-Free Control

**Abstract**—This paper introduces LOOK TRACK VISION, a multi-modal Human-Computer Interaction (HCI) system that can be used to control computers hands-free with the help of gaze direction, blink gestures, and voice commands. The system is built based on a regular webcam and a Convolutional Neural Network (CNN) model to classify gaze directions (LEFT, RIGHT, UP, DOWN, CENTER), with MediaPipe facial landmarks to estimate blink and head-pose. Multi-modal interface is an interface which incorporates gaze based cursor control, blink based clicking, voice-to-text communication and an assistive Communicator module. The solution does not require any specialized hardware to operate so that it is low-cost, portable and can be used by users with motor impairments. The proposed CNN was experimentally analyzed with the highest accuracy of up to 95% in webcam-based inputs with various lighting conditions. The elaborated desktop application also confirm the usability, robustness and potential of the system as a comprehensive HCI solution.

**Index Terms**—Computer Vision, Image Processing, Human–Computer Interaction (HCI), Convolutional Neural Network (CNN), Gaze Tracking.

## I. INTRODUCTION

Human-Computer Interaction (HCI) has progressively moved toward more natural, accessible, and intuitive ways of communication. Individuals with motor impairments face difficulties when using traditional input methods like mouse and keyboard. This leads to hands-free interaction with computer applications. With the advancements in deep learning, computer vision, gaze-based control and voice-assisted interfaces have emerged as effective alternatives for inclusive digital interaction.

Gaze tracking provides a non-invasive and device-independent way of interpreting user intent by analyzing eye movement patterns. Modern systems combine facial landmark extraction, gaze direction classification and blink detection to translate visual cues into computer commands. Moreover, these methods enable hands-free navigation and interaction with computer applications.

The existing gaze tracking systems have limitations such as high hardware cost, accuracy, sensitivity to lighting and limited multi-modal support [1] [2]. This designed HCI system introduces a real-time, low-cost and multi-modal interaction. In addition to that, the proposed system is integrated with a lightweight Convolutional Neural Network (CNN) for gaze direction estimation with MediaPipe-based iris and facial landmark detection for blink and head-pose estimation.

A structured mapping layer converts LEFT, RIGHT, UP, DOWN, CENTER gaze predictions and blink gestures into mouse control actions and also, voice integration supports text typing for users with limited physical disabilities. The

system further incorporates a calibration mechanism, multi-modal assistive tools support for desktop environments. By relying solely on consumer-grade webcams, LOOK TRACK VISION aims to provide an accessible, practical, and affordable solution that significantly enhances independence and digital inclusion for users with motor disabilities.

## II. METHODOLOGY

A CNN model was designed and trained specifically for real-time gaze direction recognition and combined with an additional landmark-based blink detection module to enable full mouse control in computer using only eye movements.

### A. Model Development

The responsibility of the trained CNN model is classifying five primary gaze directions as shown in Fig. 1.

1) *Dataset Preparation and Preprocessing*: To obtain variation in eye images, several publicly accessible datasets were collected from online sources, including the MRL eye dataset [6], Mendeley gaze dataset [3], SBVPI Sclera dataset [4], and Kayvan Shah eye dataset [5]. In addition to downloaded datasets, a custom dataset of a total of 1,012 eye images were captured from 12 different individuals under varying lighting conditions using standard laptop webcam.

Both collected images were combined, resized to 64×56 pixels, and converted to grayscale. Data augmentation techniques were applied to increase variability and improve model robustness. Augmentation techniques included normalization, random rotation, width and height shifting, shear transformation, and zoom adjustment. Finally, the total dataset was split into 80% for the training and 20% for testing. Table I shows the distribution of eye image counts per class for both the training and testing datasets.

2) *Model Design and Training*: The CNN gaze direction detection model architecture consisting of three convolutional layers with ReLU activation, each followed by a max pooling layer to extract and reduce spatial features effectively. After the final pooling layer, the feature maps are flattened and passed to a dense layer with 512 ReLU-activated neurons, followed by



Fig. 1. Primary gaze directions for model training.

TABLE I  
TRAINING AND TESTING DATASET IMAGE COUNT PER EACH CLASS

| Direction | Training Dataset (images) | Testing Dataset (images) |
|-----------|---------------------------|--------------------------|
| Center    | 3,083                     | 797                      |
| Down      | 3,119                     | 873                      |
| Left      | 3,260                     | 800                      |
| Right     | 3,555                     | 901                      |
| Up        | 3,264                     | 816                      |

a dropout layer to minimize overfitting. The output layer contains five SoftMax-activated neurons corresponding to the five gaze direction classes. The model was optimized using Adam optimizer with categorical cross-entropy loss and trained for 15 epochs. The overall architecture is shown in Fig. 5.

### B. Real-time Gaze Direction Recognition System

The real-time gaze direction recognition process integrates CNN-based gaze direction prediction, Calibration process, Blink detection and head pose estimation, enabling more precious and robust mouse control.

1) *CNN Gaze Direction Prediction*: A standard webcam is used to continuously capture the user's face. The captured video frames are processed in real time, where the eye regions are extracted, resized to the required input dimension (64x56), and converted to grayscale before being fed into the trained model. Finally, the model accurately predicts the user's gaze direction. This setup provides a low-cost, webcam only solution without requiring any specialized hardware.

2) *Calibration Process*: During the calibration process, the user is instructed to focus on four predefined points on the screen as illustrated in Fig. 3. At the end of this calibration process user is instructed to close both eyes to record the blink-based threshold values. This procedure establishes landmark-based gaze direction prediction, ensuring that normal viewing of the screen does not accidentally trigger cursor movements. The final gaze direction is determined by combining predictions from both the CNN-based and landmark-based gaze estimation.

3) *Blink Detection*: In addition to gaze direction, the blink is included as another input method. Blinks are detected using a combination of the closed-eye threshold values obtained during the calibration process and the Eye Aspect Ratio (EAR) [7] calculation using specific facial landmarks around the eye shown in Fig. 2. The EAR is calculated using (1).

$$EAR = \frac{\|P_2 - P_6\| + \|P_3 - P_5\|}{2 \times \|P_1 - P_4\|} \quad (1)$$

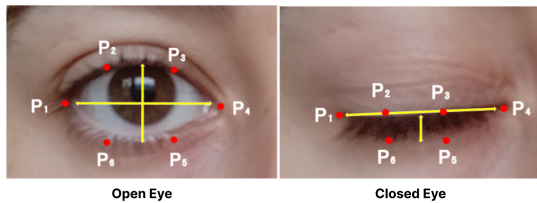


Fig. 2. Eye landmarks position for the EAR calculation.

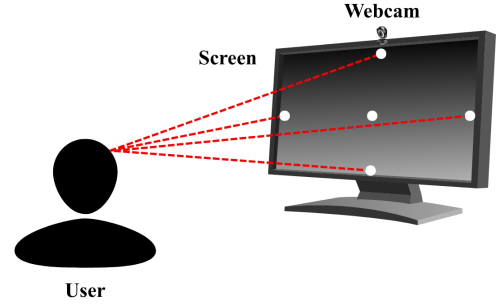


Fig. 3. Calibration setup.

Here, P1 to P6 represent six main eye landmarks. During eye open, the vertical distance between the eyelids is large compared to the horizontal width of the eye, which gives a higher EAR value. During eye blinks, the eyelids come closer together, reducing the vertical distance and resulting the EAR to drop below a defined threshold 0.2. If this drop continues for a few frames, it is detected as a blink. These left and right eye blinks are mapped to mouse left and right click events, respectively.

4) *Head Pose Estimation*: Head pose estimation is implemented to determine the user's head orientation relative to the webcam by analyzing the pitch (up/down), yaw (left/right), and roll (tilt) angles obtained from facial landmarks as shown in Fig. 4 [8].

Based on these angles, the system classifies the user's head direction as follows:

- Yaw  $< -12^\circ$  : Left
- Yaw  $> +12^\circ$  : Right
- Pitch  $< -12^\circ$  : Down
- Pitch  $> +12^\circ$  : Up
- Otherwise: Forward

This prevents misclassifications in gaze predictions caused by head movement and ensures more stable and accurate gaze-based computer control.

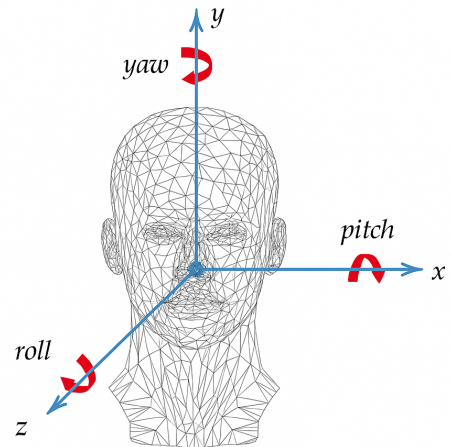


Fig. 4. Head pose estimation.

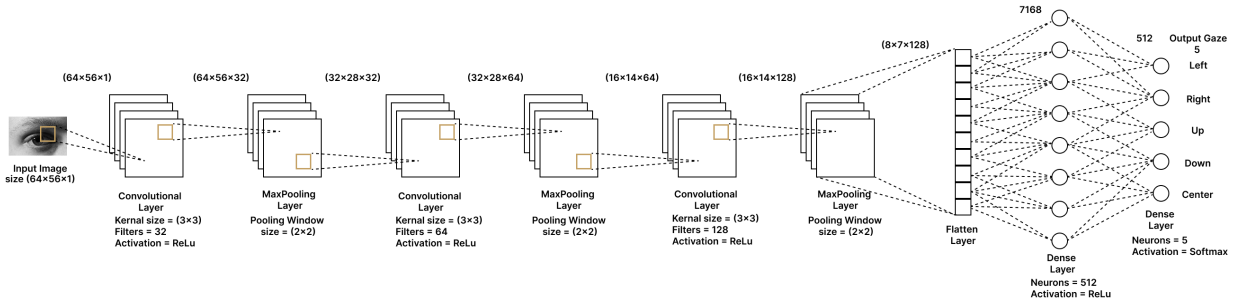


Fig. 5. CNN model architecture.

### C. Software Architecture

The unified hands free Human Computer Interaction system is developed into a modular and interactive system that incorporates the real-time gaze estimation, blink control and voice to the software application named LOOK TRACK VISION. It is a Python based system written using Tkinter, inference modal of OpenCV, media pipe and CNN which finally consists of the two large layers of architecture namely Software User Interface Layer that governs interactions, user level and navigation, and accessibility and the implementation of multi-modal commands, blink interpretation, gaze prediction, and cursor control are carried out by Core Functional Layer.

1) *Software User Interface Layer*: A user interface layer is created as an interactive layer based on Tkinter and lightweight with its clearly defined, modularized features of the Home, Calibration, Gaze Test, Tips, Communicator, Virtual Keyboard, Sokoban Game, Information and Settings. The screen creates standalone user interface modules which interact with the operations in the back-end via event listeners and common state variables. This application can be controlled with the usage of blinks as well as with the gaze directions once the application is launched.

**Home Screen** : The Home Screen is the central navigation point of LOOK TRACK VISION which offers an easy access to the primary modules of the application like Calibration, Gaze Test, Communicator, Virtual Keyboard, Game, Tips and settings as shown in Fig. 6. It has big eye catchy buttons which can be tapped via the blink gesture and this means that the

user can easily open various functionalities, particularly when the individual has low mobility. System status indicators (like camera readiness and control mode (Auto/Manual)) are also displayed on the Home interface so that the users are able to verify the system state prior to making any action.

**Calibration Screen** : The users are recommended to repeat the calibration process whenever the position and environment of the users change. So this Calibration interface takes the user through a sequence of ordered fixation prompts (left, right, up, down, and closed eyes) and countdown clockers, visual cues, and progress bars are shown to guarantee the right positioning and consistent recording of customized threshold values.

**Gaze Test screen** : This display gives the real-time visual feedback in detecting blink and classifying gaze. Therefore, this allows the user to test the functionality of the gaze direction detection and then only give the full control of the computer.

**Communicator module** : Blink gestures and directional mouse control by gaze allow users with speech impairment to construct sentences with the Communicator Module with popularly known Augmentative and Alternative Communication (AAC). The mouse pointer is controlled using gaze directions and the left blink is used as the left mouse click to activate the grid-arranged words. The vocalization of the composed sentence is done by text-to-speech (TTS) engine (pyttsx3), which allows an effective channel of communication.

**Virtual Keyboard and Sokoban Game** : The hands free typing on all applications enables the user with limited mobility to have a better lifestyle with the help of the blink and gaze directions in locating the cursor and choosing the key in the virtual keyboard. Besides this, the process of visual motor reinforcement and the involvement of the mind is also encouraged by the remodeling of the Sokoban game in the LOOK TRACK VISION that can be played by all directional gaze changes as similar as the key handling of the keyboard.

**Setting Screen** : Finally, the Settings screen also provides the user with some customization options such as camera selection, system tray behaviour, rest reminder intervals and voice feedback preferences. The rest reminder confirmations are included in the system to remind the users about the rest reminder to ensure the wellness of the users. All these settings are stored permanently in the system to guarantee the same behaviour throughout the sessions. A lightweight Instruction

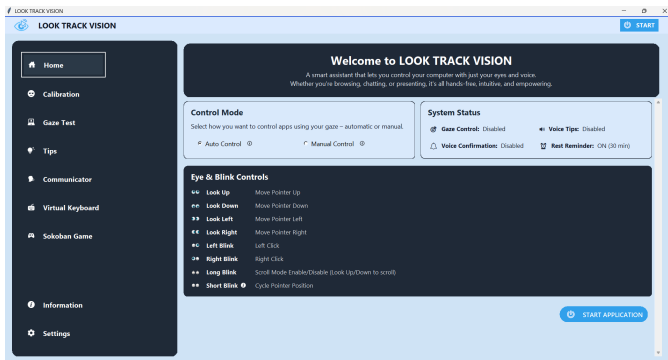


Fig. 6. Home screen of the LOOK TRACK VISION desktop application.

Tray Bar remains active all the time when the main window is minimized to ensure the much needed system status and required gaze gestures are always accessible.

2) *Core Functional Workflow*: The multimodal interaction and real-time visual processing make the core functional layer of the system. The MediaPipe FaceMesh is applied at the beginning of the workflow in order to extract face and eye landmarks using continuous webcam input. The eye regions are then distinguished and categorized with the trained CNN as either left, right, up, down or center. This direction is eventually translated into on-screen cursor movement with the OS-level mouse control via the PyAutoGUI and continuous pointer movement can be achieved.

Another control channel is the blink detection. Short dual-eye blinks are plotted with repositioning the cyclic cursor at predefined areas of the screen and controlled left-eye and right-eye blink actions identified and executed left-click and right-click actions with eye-lid points. The scrolling and cursor mode switching are activated with the help of the long dual blinks. In this way the hybrid gaze-blink system will offer a complete mouse interaction model without the need to have any physical input devices.

In addition to that, two operating modes are provided by the system to ensure convenience and flexibility:

**a) Auto Mode:** Gaze based mouse control is activated default when the application starts. The system initiates inference of the gaze direction and blink gestures immediately and therefore this mode is the best in continuity of hands-free use especially for the users who having motor impairments.

**b) Manual Mode:** The gaze based mouse control is activated in this mode after voice command is given that is "Start Gaze Control". This gets rid of accidental movement of the cursor and provides greater control in a shared or multi-user environment as illustrated in the Fig. 9.

The system has a voice interaction component containing text-to-speech (pyttsx3) output and offline speech recognition of Whisper to improve usability. This integration makes the hands free text dictation, instant audio confirmation feedbacks and gaze based mouse controlling to start and stop possible in the manual mode. The integration of the gaze direction, the blink gestures, and speech commands creates a very powerful multi-modal interaction paradigm and these features are provided with the power of simple, easy, and totally hands-free computer usage.

### III. RESULTS AND EVALUATION

#### A. Model Training and Validation Loss Over Epochs

The training and validation loss curves in Fig. 7, show a steady decrease over the training process, indicating effective learning and stable optimization. Both losses drop rapidly during the initial epochs before gradually flattening, suggesting the model approaches convergence. The validation loss consistently remains slightly lower than the training loss. It implies strong generalization and no signs of overfitting. Overall, the loss behavior demonstrates that the CNN was

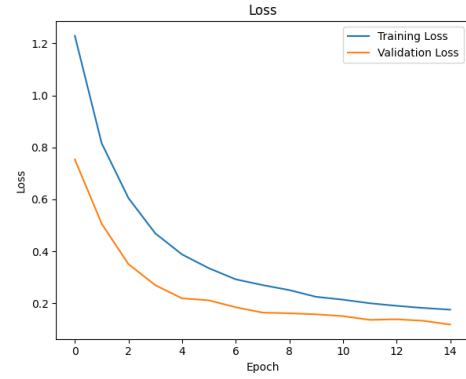


Fig. 7. Model training and validation loss over epochs.

trained efficiently, achieving stable performance and reliable predictive capability.

#### B. Model Training and Validation Accuracy Over Epochs

The training and validation accuracy curves in Fig. 8 show a clear upward trend across epochs, indicating that the model continuously improves its predictive capability as training progresses. Validation accuracy remains slightly higher than training accuracy throughout, suggesting strong generalization to unseen data and the absence of overfitting. Both curves gradually converge to values above 0.90, demonstrating that the CNN achieves high performance on both training and validation sets. Overall, the accuracy plot confirms that the model learns effectively, stabilizes well, and maintains reliable classification performance across all gaze-direction classes with 95% accuracy.

#### C. Class-Wise Evaluation and Confusion Matrix Analysis

Fig. 10, demonstrates strong predictive performance across all five gaze-direction classes. The model shows clear diagonal dominance, indicating that most samples were correctly classified, with only minor confusion between visually similar classes such as center and up/left. The right class shows particularly strong separability, with almost no misclassifications. These findings are consistent with the high precision, recall, and F1-scores reported for each class.

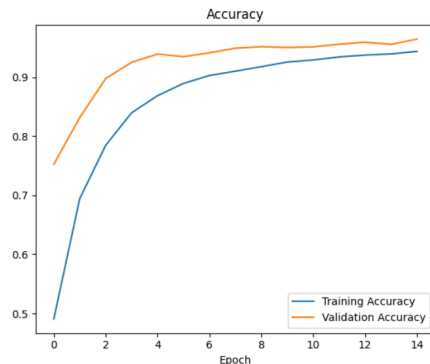


Fig. 8. Model training and validation accuracy over epochs.

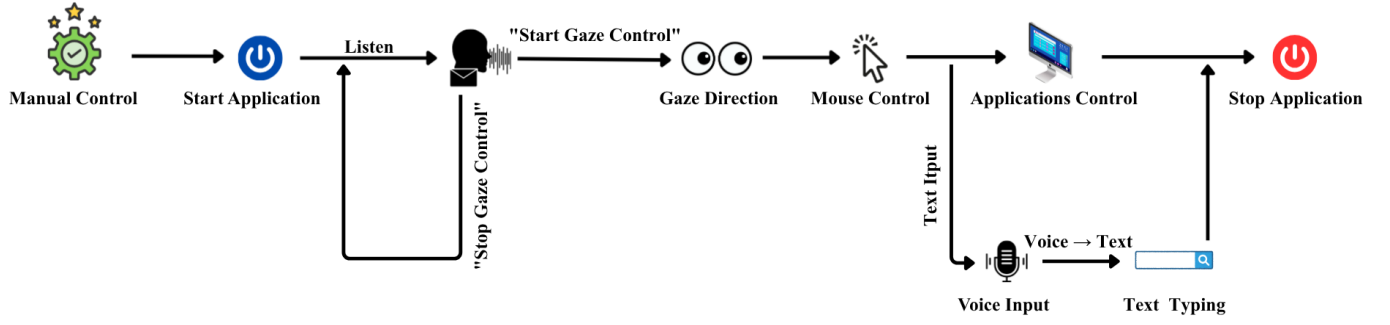


Fig. 9. Work flow of manual control mode.

#### D. Performance Under Different Lighting Conditions

Across multiple users, the CNN consistently produced accurate LEFT, RIGHT, UP, DOWN, and CENTER predictions. The system gives correct predictions even when illumination levels varied. In brighter environments, iris contrast and landmark visibility were higher, resulting in highly stable outputs, while in low-light settings the MediaPipe landmark detector continued to function reliably, allowing the model to preserve its classification accuracy. Although minor variations occurred due to reduced contrast under dim lighting. But no significant degradation in performance was observed. Overall, the results confirm that the system is robust to common indoor lighting variations and capable of delivering reliable gaze estimation in typical real-world settings.

#### E. Blink Detection Performance

The system's blink detection mechanism is based on EAR combined with MediaPipe eyelid landmark. The model successfully distinguished between left-eye blinks, right-eye blinks, normal blinks, and long blinks after calibration, enabling accurate mapping to interaction commands such as left-click, right-click, scrolling, and cursor repositioning. Blink recognition remained stable even with natural blink variability and minor lighting differences, and the calibrated thresholds

minimized false detections. This consistent performance confirms that blink-based input is a robust and intuitive complement to gaze-direction control, enhancing the overall usability of the system for hands-free interaction.

#### F. Head-Pose Behavior and System Stability

The system is intentionally designed to suppress gaze predictions whenever the user's head deviates from a forward-facing position. During testing, the head-pose module correctly identified LEFT, RIGHT, UP, and DOWN head orientations in all scenarios, and the gaze-prediction window appropriately displayed no output to avoid misclassifications caused by distorted or partially obstructed eye regions. Once the user returned to a forward orientation, predictions resumed instantly and accurately. This behavior significantly improves reliability by ensuring the gaze classifier operates only when the eye region is clearly visible. The results confirm that forward head pose is a necessary condition for stable performance and that the incorporated head-pose logic enhances overall system robustness and user safety.

### IV. DISCUSSION

The evaluation of the system demonstrates that multi-modal, webcam-based HCI is both effective and practical for hands-free computer control. The CNN based gaze-direction model, has an accuracy of 95% approximately. Using the standard web-cam based images, the five gaze-directions have been classified namely LEFT, RIGHT, UP, DOWN and CENTER. Moreover, the system does not rely on infrared illumination or proprietary eye-tracking sensors. As a result, it makes the solution significantly more affordable and widely deployable.

MediaPipe facial landmarks further strengthened robustness for stable gaze estimation. It enables precise eye-region extraction and reliable head-pose analysis. Blink detection is implemented through EAR-based thresholding mechanism, accurately differentiated between right, left, normal and long blinks. These blinks are mapped to common actions such as clicking and scrolling. Therefore, it creates a smooth and intuitive interaction experience that complemented the gaze-based pointer control.

On the other hand, several limitations were observed. Extreme lighting conditions, reflective eyeglasses and sudden

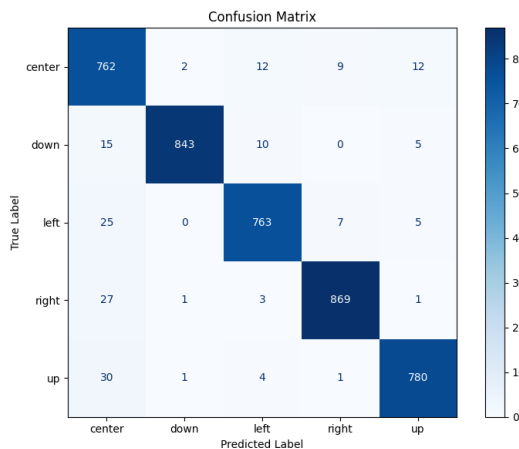


Fig. 10. Confusion matrix of the trained CNN model.

head movements are some of them. Therefore, the landmark visibility will be affected and the accuracy will be reduced. Although head-pose suppression prevents misclassifications, environmental factors can influence the performance. In addition, prolonged gaze usage may lead to eye fatigue. Multitasking across multiple real-time threads may introduce latency on low-end hardware. Despite these constraints, the system demonstrates a reliable and accessible gaze-based interaction platform, specially for the individuals with physical disabilities.

## V. FUTURE WORK

Several enhancements can further improve its robustness, usability and adaptability to real-world environments. In addition to the primary gaze directions add the additional classes such as diagonal gaze directions to expand the wider range of gaze-based input and improve the system interaction precision and this can be expanded to the mobile platforms to improve portability and accessibility. The model can also be further trained to increase the accuracy and robustness across diverse lightning conditions, slight head movements and eyeglass reflections. Another valuable extension is integrating the real time gaze-based control techniques with IoT and smart-home devices to interact with daily-use systems more seamlessly. Moreover, AI-based eye-health analytics, such as monitoring the blink rate, screen time, and visual strain, may contribute to the safe use of AI in the long-term.

## VI. CONCLUSION

The development of the LOOK TRACK VISION system shows that the combination of gaze tracking, blink recognition, and voice interaction can be successfully implemented to provide a fully hands-free HCI environment using only low-cost, webcam-based hardware. With an accuracy of approximately 95%, the CNN gaze model, MediaPipe-based iris, head-pose estimation and an EAR-based blink classifier enabled reliable and natural control of cursor movement and selection actions. The inclusion of communication tools, such as the Communicator module, virtual keyboard, and additional multi-modal features, further enhanced accessibility for users with physical disabilities. Although the system is affected by lighting variations, reflective eyeglasses, and occasional latency on low-end devices, the results demonstrate that affordable deep-learning and computer-vision techniques can significantly improve independence, usability, and digital inclusion. Overall, LOOK TRACK VISION provides a strong foundation for future developments in adaptive gaze-controlled systems and inclusive HCI technologies.

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